ASSESSMENT TOOL - 1

Course code	CSA0702	Course Title	Computer Networks
Co Assessed	Co 1	Assessment	Tool - Short Answers
Weightage		Total marks	20 marks

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Section : Batch	IT - 2024	Date of Submission:	17/07/2026
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Total marks	20	Obtained marks	17

ASSESSMENT TOOL-1

SHORT ANSWERS

Name : J. Kirthiga

Reg No: 192421265

Course name: computer Networks

Course code : CSA0702

1. Transport Layer + Data link layer

Given

file size = 150 MB

Bandwidth = 50 Mbps

Segment size = 1500 bytes

Header + Trailer = 40 bytes/frame

Step 1 : No. of. Segments

Using 1MB = 1,000,000 bytes 150 MB = 150,000,000 bytes

$$\text{Segments} = \frac{150,000,000}{1500} = 100,000$$

Segments = 100,000

Step 2 No. of. frames

Each Segment becomes one frame

frames = 100,000

Step 3 Total overhead

$100,000 \times 40 = 4,000,000 \text{ bytes} = 4MB$

Step 4 Transmission time

150 MB = 1200 Mb

$$\frac{1200}{50} = 24 \text{ sec}$$

Transmission time = 24 Seconds

2) Sliding Window

Given

Window size = 6

Total frames = 48

Frame size = 1024 bytes

Step 1

$\frac{48}{6} = 8$ Transmission windows = 8

Step 2

one frame lost in every window

Retransmissions = $8 \times 1 = 8$

Theory:
flow control prevents the Sender from overwhelming the receiver, reducing congestion & Improving effisency

3) IP Fragmentation

Packet = 12,000 bytes

MTU = 1500 bytes

Header = 20 bytes

Step 1

Each fragment can carry

$1500 - 20 = 1480$ bytes

Step 2

$\frac{12000}{1480} = 8.1$

Need 9 fragments

Step 3 Header overhead

$9 \times 20 = 180$ bytes

Theory:

The Network layer fragments large packets and ressembles them at the destination using Identifica-tion numbers and fragment offsets.

4) Sliding Window.

Transmission windows = $48/6 = 8$

Retransmissions = 8

Theory:

flow control regulates transmission speed and avoids receiver buffer overflow

5) Session layer:

Given

Video = 600 MB

Checkpoint every 60 MB

failure after 390 MB

Step 1

Checkpoints before failure

60, 120, 180, 240, 300, 360

Total checkpoints = 6

Step 2

Recovery starts from 360 MB

Retransmitted data

$390 - 360 = 30$ MB

Retransmission = 30 MB

Theory:
Synchronization checkpoints allow communication to resume from the last checkpoint instead of transmitting the entire file

6. Presentation layer:

Given

original = 900 MB

compression = 40%

Encryption overhead = 5%

Step 1

compressed size

$900 \times 0.60 = 540$ MB

Step 2 Encrypted size

$540 \times 1.05 = 567$ MB

Step 3 Reduction

$900 - 567 = 333$ MB

$\frac{333}{900} \times 100 = 37\%$

Theory:
The presentation layer compresses data to reduce transmission size and encrypts it for confidentiality

7) FTP Application

Given

Data = 3 GB

Throughput = 120 Mbps

Request size = 3 MB

Step1

3 GB = 3000 MB

Requests

$3000/3 = 1000$

Request = 1000

Step2

3 GB = 24000 Mb

Transmission time

$$\frac{24000}{120} = 200\,sec$$

Theory:

The Application layer provides services like FTP file, transfer, authentication and user interaction.

8) End to End Delay

Given

Application = 4 ms

Transport = 3 ms

Network = 5 ms

Data link = 2 ms

physical = ~~8~~ 1 ms

propagation = 18 ms

Transmission = 12 ms

Total = 4 + 3 + 5 + 2 + 1 + 18 + 12 = 45 ms

Total delay = 45 ms

Theory:

Each OSI layer performs specific processing before data is transmitted propagation and transmission delays contribute to the overall communication time

9) protocol overhead

Useful data = 80 MB

frame Size = 1600 bytes

Overhead = 80 bytes

Step 1

frames

$80{,}000{,}000 / 1600 = 50{,}000$

Step 2

Total overhead

$50{,}000 \times 80 = 4{,}000{,}000$ bytes $= 4$ MB

Step 3

Efficiency

Useful playload / frame

$1600 - 80 = 1520$

$\frac{1520}{1600} \times 100 = 95\%$

Theory:

Higher protocol overhead reduces bandwidth efficiency because more bytes are used for headers and trailers instead of user data.

10) Multiple OSI layers

Given

Original file = 240 MB

Compression = 30 %

Segment Size = 1400 bytes

frame overhead = 60 bytes

Step 1

compressed size = 240 × 0.70 = 168 MB

Step 2

compressed bytes

168,000,000

Segments

$\frac{168,000,000}{1400} = 120,000$

Segments = 120,000

Step 3

protocol overhead

120,000 × 60 = 7,200,000 bytes

= 7.2 MB

Step 4

Final transmitted Data

168 + 7.2 = 175.2 MB

Theory:

Presentation layer : compresses & encrypts data

Transport layer : Segments data and ensures reliable delivery

Data link layer: Adds frame headers/ trailers for error detection

physical Layer : Transmits the final bits over the communication medium.

Understanding of concepts	Explanation	Application	Organization	Accuracy	Clarity.
25%.	25%.	20%.	15%.	10%.	5%.
5	5	4	2	0.5	0.5